Question No : 4 of 31

Marks: 1 (Time 1 Min)

The set of all left (or right) cosets of a subgroup H of a group G does not form partition of G.

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Answer

- ○ True
- ◉ False ✓

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Q.29: Shaw that $(S_3, \circ)$ is a non-abelein Joup.

Sol.

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Here $(1\ 2)(2\ 3) = (1\ 2\ 3)$

And $(2\ 3)(1\ 2) = (1\ 3\ 2)$

Thus $(1\ 2)$ and $(2\ 3)$ do not commute. Since these elements belong

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In general, the set of all $n \times n$ matrices is ---------- under matrix multiplication.

Answer

- a group
- not a group ✔
- not a semigroup
- abelian

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Orbits

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Definition

An orbit of a permutation p is an equivalence class under the

$a \sim b \Leftrightarrow b = p^n(a)$,

for some n in $\mathbb{Z}$.

ENG

Every group is isomorphic to ---------- group.

Answer

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- a permutation
- an abelian
- a cyclic
- Dihedral

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8 of 16 — Fit to page — Page view — Read aloud — Add notes

The subset of $\mathbb{Z}$ consisting of all integers that are multiples of both r and s is ---------- group generated by the least common multiple of two positive integers r and s.

Answer

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- (•) cyclic ✔
- ( ) non-cyclic
- ( ) non abelian
- ( ) commutative

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Question No : 27 of 31

Marks: 1 (Time 1 Min)

Let $f: G \to H$ be a homomorphism, then $f(e_G) = ---$ *for* e_G *and* e_H *are identities of* G *and* h *respectively.*

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Answer

- $e_H \cap e_G$
- e_G ✓
- e_H

Let $H = \{(2t, t) | t \in R\}$ be a subgroup of R^2, then cosets of H in R^2 are lines ___ to the line $y = \frac{x}{2}$.

Answer

- same
- perpendicular
- parallel ✓
- intersecting

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Question No : 30 of 31

Show that $(S_3, \cdot)$ is a non-abelian group.

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Answer

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Which of the following is NOT a rational number?

Answer

- 5
- $\frac{22}{7}$
- 0
- π

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Question No : 30 of 31

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Marks:

Show that $(S_3, \circ)$ is a non-abelian group.

Answer

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Consider the permutation $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 1 & 4 \end{pmatrix}$. Which of the following is false?

Answer

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- $\alpha(2) = 3$
- $\alpha(3) = 1$
- $\alpha(1) = 2$
- $\alpha(4) = 3$

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Here $(1\ 2)(2\ 3) = (1\ 2\ 3)$

And $(2\ 3)(1\ 2) = (1\ 3\ 2)$

Thus $(1\ 2)$ and $(2\ 3)$ do not commute. Since these elements belong to S_n for $n \geq 3$, it follows that S_n is non-abelian for $n \geq 3$.

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Question No : 14 of 31

Marks: 1 (Time 1 Min)

Every permutation can be written as a ---------- of transpositions.

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Answer

- Sum
- Product
- Quotient

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$\phi : (R^+, .)$

step 1: $(R, +) = \phi(x) = e^x$

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2: $(R^+, .) = \phi(y) = e^y$

3: $\alpha \in R \quad \ln(\alpha) \in R$

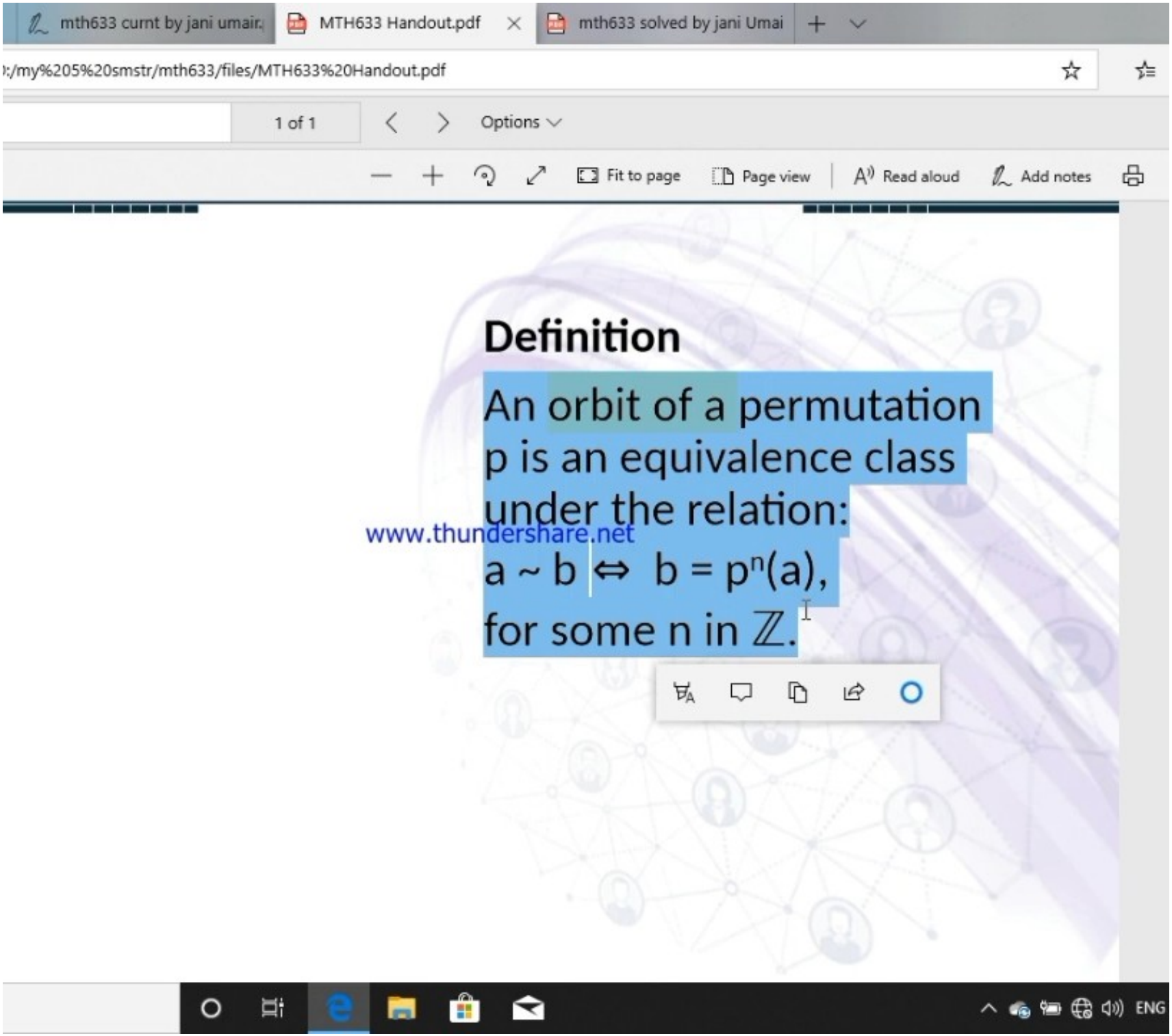
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The group H is a subgroup of group G. If order of G is 18, then order of H can be ----------

Answer

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- 5
- 7
- 11
- 9 ✓


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Definition
An orbit of a permutation p is an equivalence class under the relation:
a ~ b ⇔ b = p^n(a),
for some n in ℤ.
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50

Question No : 20 of 31

Marks: 1 (Time 1 Min)

Let $\phi : G \to H$ be an isomorphism, then if G is abelian, then H is ----------

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Answer

- ○ cyclic
- ◉ abelian ✓
- ○ Non-commutative

A subset H of a group G is a subgroup of G if and only if H is _____ under the binary operation $*$ of G, and for all $a \in H$ it is true that _____.

Answer

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- closed, $a^{-1} \in H$ ✓
- closed, $b^{-1} \in H$
- associative, $a^{-1} \in H$
- associative, x^{-1}, y

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59

Question No : 1 of 31

Marks: 1 (Time 1 Min)

The order of the group $\{1, -1, i, -i\}$ is ----------

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Answer

- 2
- 3
- 4 ✔

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Cosets of $H=\{(2t,t) \mid t \in \mathbb{R}\}$

$(a,b) + H = \{(a+2t,b+t)\}$

Set $x = a+2t$, $y = b+t$ and eliminate t:

$y = b + (x-a)/2$

The subgroup H is the line $y = x/2$.

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The cosets are lines parallel to $y = x/2$!

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The function $f: X \rightarrow Y$ is onto if ----------

Answer

- $f[X] = X$

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- $f[Y] = X$
- $f[Y] = Y$
- $f[X] = Y$ ✓

Question No : 28 of 31

Marks: 3

Write down the orders of all subgroups of a group of order 12

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Answer

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$\langle R-\{0\}, \cdot \rangle$ is ---------- of $\langle R, \cdot \rangle$

Answer

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- (●) subgroup ✓
- ( ) not subgroup
- ( ) trivial subgroup
- ( ) non-trivial subgroup

$\phi : (R, +) \quad \phi(x) = e^x$

Isomorphic Binary operation:-

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$\phi : (R^+, .)$

step 1. $(R, +) = \phi(x) = e^x$

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Question No : 11 of 31

Marks: 1 (Time 1 Min)

Every subgroup of a cyclic group is ---------- group.

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Answer

- (•) Cyclic
- ( ) Non- cyclic
- ( ) Non-abelian

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15 of 16 Fit to page Page view Read aloud Add notes

37

Question No : 31 of 31

Marks:

Show that the $\varphi: R \rightarrow R^{+}$ defined by $\varphi(x) = e^{x}$ is the isomorphic binary structure from $\langle R, + \rangle$ to $\langle R, - \rangle$.

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Answer

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Cycle Decomposition

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Theorem

Every permutation can be written as a product of transpositions.

Proof

Use the lemma plus the previous theorem.

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2: $(R^+, .) = \phi(y) = e^y$

3: $x \in R \quad \ln(x) \in R$

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$\phi(\ln x) = e^{\ln x}$.

Proove.

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Let H be a subgroup of G and a, b are in G, then aH and bH are ----------.

Answer

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- equal only
- disjoint only
- either equal or disjoint ✓
- none of these

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The order of the group $\{1, -1, i, -i\}$ is ----------

Answer

- 2
- 3
- 4 ✔
- 5

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Question No : 2 of 21 Marks: 1 (Time 1 Min)

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Question No : 21 of 31

Marks: 1 (Time 1 Min)

Let ϕ be a homomorphism from a group G into a group G'. If H is a subgroup of G, then $\phi[H]$ is a subgroup of _____.

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Answer

- G ✓
- G'
- H

Which of the following is the cyclic group?

Answer

- (●) Group of order 6 ✓
- ( ) Group of order 10
- ( ) Group of order 12
- ( ) Group of order 17

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For each positive divisor k of n, the group $\langle a \rangle$ has exactly one subgroup of order k, namely _____.

Answer

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- $\langle a^{n/k} \rangle$ ✓
- $\langle a \rangle$
- $\langle a^k \rangle$

Let $H = \{(2t, t) | t \in R\}$ be a subgroup of R^2, then cosets of H in R^2 are lines ___ to the line $y = \frac{x}{2}$.

Answer

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- same
- perpendicular
- parallel ✓
- intersecting

The function $f: X \rightarrow Y$ is onto if ----------

Answer

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- $f[X] = X$
- $f[Y] = X$
- $f[Y] = Y$

39

Question No : 29 of 31

Marks: 3 (Time 6 Min)

Write three properties of real numbers.

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Answer

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There aretl:ee properties of resl number are:

1. Clousr property
2. Associate proprtry
3. Cummative property
4. Indentity property



The binary structures $\langle Q, + \rangle$ and $\langle R, + \rangle$ are not isomorphic because ----------

Answer

- their binary operations are different.

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- their cardinalities are not same. ✓
- commutative property holds in Q only.
- commutative property holds in R only.

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Question No : 5 of 31

Marks: 1 (Time 1 Min)

The set of all left (or right) cosets of a subgroup H of a group G has ----------number of elements as H.

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Answer

- (•) the same ✓
- ( ) greater
- ( ) lesser

37

Question No : 31 of 31

Show that the $\varphi: R \rightarrow R^+$ defined by $\varphi(x) = e^x$ is the isomorphic binary structure from $\langle R, + \rangle$ to $\langle R, - \rangle$.

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Answer

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Question No : 10 of 31

Every ---------- group is abelian.

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Answer

- (•) Cyclic
- ( ) Non- cyclic
- ( ) Non-abelian
- ( ) Finite

Question No : 25 of 31

Marks: 1 (Time 1 Min)

An orbit of a permutation p is ––– class under the relation: $a \sim b \Leftrightarrow b = p^n(a)$, for some n in Z.

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Answer

- (•) an equivalence ✓
- ( ) a conjugacy
- ( ) simple
- ( ) [illegible]

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Question No : 14 of 31

Marks: 1 (Time 1 Min)

Every permutation can be written as a ---------- of transpositions.

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Answer

- Sum
- Product
- Quotient

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Group	order 1	order 2	order 3	order 4	order 6	order 12
$Z/(12)$	1	1	2	2	2	4
$Z/(2)^2 \times Z/(3)$	1	3	2	0	6	0
A_4	1	3	8	0	0	0
D_6	1	7	2	0	2	0
$Z/(3) \times Z(4)$	1	1	2	6	2	0

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$\prod_{i=1}^{n} G_i$ is the ---------- notation for all the group operations for $G_1, \ldots, G_n$.

Answer

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- (●) additive
- ( ) multiplicative ✔
- ( ) subtractive
- ( ) division

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Cosets of $H=\{(2t,t) \mid t \in \mathbb{R}\}$

$(a,b) + H = \{(a+2t,b+t)\}$

Set $x = a+2t$, $y = b+t$ and eliminate t:

$y = b + (x-a)/2$

The subgroup H is the line $y = x/2$.

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The cosets are lines parallel to $y = x/2$!

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The group H is a subgroup of group G. If order of G is 18, then order of H can be ----------

Answer

- 5
- 7
- 11
- 9

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48

Question No : 22 of 31

Marks: 1 (Time 1 Min)

If N is a normal subgroup of a group G, the left cosets of N in G are ---------- as the right cosets of N in G.

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Answer

- not the same
- the same
- partially same

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Let $f: G \to H$ be a homomorphism, then $f(e_G) = ---$ *for* e_G *and* e_H *are identities of* G *and* h *respectively.*

Answer

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- $e_H \cap e_G$
- e_G ✓
- e_H
- $e_H \cup e_G$

The order of the group $\{1, -1, i, -i\}$ is ----------

Answer

- 2
- 3
- 4 ✔
- 5

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An orbit of a permutation p is $---$ class under the relation: $a \sim b \Leftrightarrow b = p^n(a)$, for some n in Z.

Answer

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- an equivalence ✓
- a conjugacy
- simple
- none of these

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